

JEE MAIN REVISION MASTER: ATOMS AND NUCLEI

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 47 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Nuclear Binding Energy & Law of Radioactive Decay	26 PYQs	Nuclear Physics
Very High	Bohr Atom Model & Hydrogen Spectral Series	21 PYQs	Atomic Physics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Nuclear Binding Energy & Law of Radioactive Decay	$N(t) = N_0 \left(\frac{1}{2} \right)^{t/T_{1/2}}, \quad T_{1/2} = \frac{\ln 2}{\lambda}$ $\approx \frac{0.693}{\lambda}, \quad R = R_0 A^{1/3}$
Bohr Atom Model & Hydrogen Spectral Series	$r_n \propto n^2, \quad E_n = -\frac{13.6}{n^2} \text{ eV}, \quad \frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right), \quad \lambda_{\text{limit}} = \frac{n_1^2}{R}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Nuclear Binding Energy & Law of Radioactive Decay (Very High Yield — 26 PYQs)

Core Principle: Nuclear radius $R = R_0 A^{1/3}$ implies nuclear density is constant and independent of mass number A . Mass defect Δm , Binding energy $BE = \Delta m \times 931.5 \text{ MeV}$. Maximum $BE/A \approx 8.8 \text{ MeV}$ around Iron (^{56}Fe). Radioactive decay law: $N = N_0 e^{-\lambda t} = N_0 (1/2)^{t/T_{1/2}}$. Half-life $T_{1/2} = (0.693 / \lambda)$, Mean life $\tau = 1/\lambda = 1.44 T_{1/2}$.

■ **Caution / JEE Pitfall:** Nuclear density is of the order of 10^{17} kg/m^3 and is IDENTICAL for all nuclei, from hydrogen to uranium!

■ **Speed Trick:** Fraction remaining after n half-lives is $(1/2)^n$. After 3 half-lives, $1/8$ remains (87.5% has decayed).

Representative Source PYQs:

- [2025 Shift] Q#2 (MCQ): "Which of the following statements is incorrect with reference of 'Nuclear force'?"... [Ans: N/A]
- [2024 Shift] Q#1 (MCQ): "The ratio of volume of Al^{27} nucleus to its surface area is (Given, $R_0 = 1.2 \times 10^{-15} \text{ m}$)..." [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "Consider the nuclear fission reaction ${}_0^1\text{n} + {}_{92}^{235}\text{U} \rightarrow {}_{56}^{144}\text{Ba} + {}_{36}^{89}\text{Kr} + 3{}_0^1\text{n}$. Assuming all the kine..." [Ans: N/A]

3.2 Bohr Atom Model & Hydrogen Spectral Series (Very High Yield — 21 PYQs)

Core Principle: Bohr's postulates: $mvr = (nh / 2\pi)$. Orbital radius $r_n \propto n^2/Z$. Energy levels $E_n = -(13.6 Z^2 / n^2)$ eV. Spectral series: Lyman ($n_1=1$, UV region), Balmer ($n_1=2$, Visible region), Paschen ($n_1=3$, Infrared). Longest wavelength corresponds to transition from $n_2 = n_1+1$; shortest wavelength corresponds to $n_2 = \infty$ (series limit).

■ **Caution / JEE Pitfall:** Only the Balmer series falls in the VISIBLE spectrum; Lyman is strictly ultraviolet, and Paschen/Brackett/Pfund are infrared.

■ **Speed Trick:** Ratio of maximum to minimum wavelength in Lyman series: $\lambda_{\max} / \lambda_{\min} = (1/(1 - 1/4)) / (1/(1 - 0)) = (4/3)$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "An electron transition takes place from excited state to ground state in hydrogen atom, then..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "Bohr's second postulate implies quantisation of..." [Ans: N/A]
- [2026 Shift] Q#3 (MCQ): "The radius of first orbit in hydrogen atom is 5.3×10^{-11} m. The kinetic energy E_K , potential energy E_P and total energy..." [Ans: B]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

An electron transition takes place from excited state to ground state in hydrogen atom, then

- (A) Its kinetic energy increases but potential energy and total energy decrease
- (B) Its kinetic energy, potential energy and total energy decrease
- (C) Kinetic energy decreases, potential energy increases but total energy remains same
- (D) Kinetic energy and total energy decrease but potential energy increases

Correct Answer: None

Solution Analysis: For hydrogen atom in the Bohr model, $E_n = -\frac{13.6n^2}{n^2}$ eV where E_n is the total energy of the electron in the n^{th} orbit. When the electron comes from an excited state to the ground state, n decreases. For example, from $n=2$ to $n=1$: $E_2 = -3.4$ eV, $E_1 = -13.6$ eV. So the total energy becomes more negative. Hence, total energy decreases. For an electron in a hydrogen atom, $K = -E$ and $U = 2E$ where K is kinetic energy and U is potential energy. Since total energy E decreases and becomes more negative: Kinetic energy $K = -E$ increases. Potential energy $U = 2E$ decreases.

Benchmark Problem 2: KCET 2025 (Q#2)

Which of the following statements is incorrect with reference of 'Nuclear force'?

- (A) Nuclear force becomes attractive for nucleon distances larger than 0.8 fm
- (B) Nuclear force becomes repulsive for nucleon distances less than 0.8 fm
- (C) Nuclear force is always attractive
- (D) Potential energy is minimum, if the separation between the nucleons is 0.8 fm

Correct Answer: None

Solution Analysis: The incorrect statement is Option C. • At very short separations ($r < 0.8$ fm) the nuclear force is strongly repulsive (a "hard core"). • For distances $r \approx 0.8-3$ fm it becomes attractive. • The potential energy reaches its minimum (stable equilibrium) at roughly $r = 0.8$ fm. Since the force is not always attractive (it's repulsive at $r < 0.8$ fm), saying "nuclear force is always attractive" is wrong.